
When Does a Speech Model Know to Hand Off? Reading Competence from Hidden States for Real-Time Escalation

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 A local speech model can keep a conversation going, and a stronger expert can
2 take over the hard questions. Calling the expert on every turn adds cost and waiting
3 time, so the local model needs an early estimate of whether its own answer will fail.
4 Hidden-state probes give such estimates for text models. It is not known whether
5 this signal survives speech and duplex fine-tuning, audio input, and the timing of a
6 native listen/speak protocol. We study these questions on frozen MiniCPM-o 4.5.
7 We fit linear failure probes on stored hidden states, audit them across layers, token
8 positions, and input modalities, and evaluate a separately calibrated answer-onset
9 gate in the model’s native protocol. The read location matters: with text input, a
10 final-layer, last-token probe inverts on held-out math queries (AUC 0.372), and the
11 same read at layer 22 reaches 0.931. A text-calibrated layer-22 probe also trans-
12 fers to speech-input states (AUC 0.855). In native sessions on 240 internal test
13 queries, the gate raises answer-content accuracy from 40.8% to 62.9% at a 51.7%
14 escalation rate, 6.8 points above the reported random-mixture reference. On four
15 external English speech-QA pools, the aggressive tier raises macro accuracy from
16 61.5% to 82.9%, against 77.1% for the reference. The cost is time: on the inter-
17 nal set, mean reconstructed time to first audio rises from 5.7 s locally to 14.4 s at
18 the aggressive tier, with a P95 of 60.8 s. The findings support mid-layer compe-
19 tence readout as a routing signal for speech models and identify the calibration
20 and serving choices that matter for real-time use.

21 1 Introduction

22 Spoken interaction leaves little time to choose a response [Stivers et al., 2009]. One practical design
23 pairs a local speech model, which keeps the conversation going, with a stronger expert that handles
24 the hard questions. Calling the expert on every turn adds cost and waiting time. Selective escalation
25 needs something else: an early estimate of whether the local answer will fail, made before that
26 answer is produced.

27 Prior work gives two starting points. For text models, hidden-state probes and verbal self-evaluation
28 can predict whether an answer is correct [Kadavath et al., 2022, Azaria and Mitchell, 2023, Kossen
29 et al., 2024], and intermediate layers can carry information that the output does not show [Orgad
30 et al., 2025]. Text routers use such signals to divide queries between models of different cost [Chen
31 et al., 2023, Ong et al., 2024, Varshney et al., 2026]. For speech, native duplex models decide for
32 themselves when to listen and when to speak [Défossez et al., 2024, Cui et al., 2026].

33 Three things remain unknown. First, it is not known whether a failure probe still works when
34 the query pool changes; a high in-distribution AUC can come from pool identity rather than from

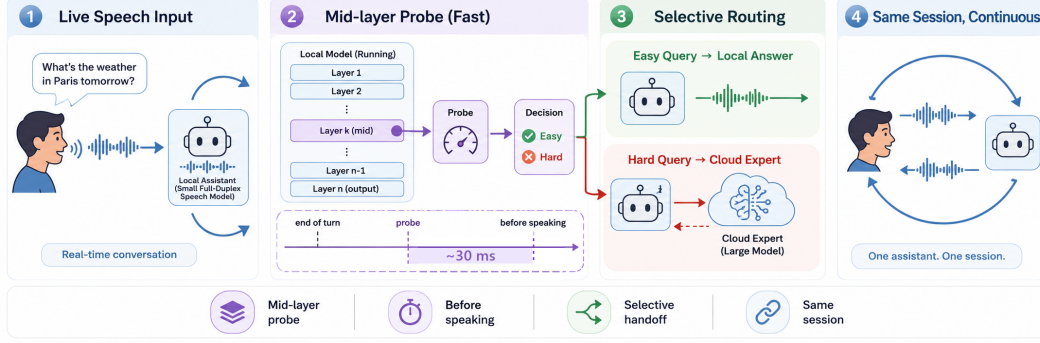


Figure 1: **The deployed system.** (1) Speech streams in 1 s chunks into a full-duplex local model that decides for itself when to speak. (2) At that speak commit, a mid-network (L22) linear read scores the turn in ~ 30 ms, before the answer is produced. (3) Lower-risk turns stay local; higher-risk turns are sent to a cloud expert while the talker stalls. (4) The returned answer is spoken verbatim in the talker’s own voice. Delivered accuracy on the internal test set rises from 40.8% to 62.9% at 52% escalation in native full-duplex sessions.

35 competence. Second, it is not known whether a probe fit on text input still reads speech-input states
 36 after speech and duplex fine-tuning. Third, it is not known where in a streaming protocol the hidden
 37 state is actually available for a routing decision. A duplex model already decides when to speak,
 38 but this floor-management decision need not reveal whether it knows the answer. It is reasonable
 39 to expect that some competence information survives fine-tuning, because the backbone carried it
 40 before. The layer, position, and modality at which it can be read have not been measured on a duplex
 41 speech model.

42 This gap matters in practice. A gate placed at the wrong layer can look good on its calibration pool
 43 and then invert on a new one, so the expert is called on the wrong turns. A probe that fails on audio
 44 input cannot be used in a speech system at all. And a read that only becomes available after the
 45 answer is generated removes the early decision that motivated escalation in the first place.

46 We study these questions on frozen MiniCPM-o 4.5 [Cui et al., 2026], with raw-backbone and
 47 other speech-model controls. Our contributions are three. First, a layer and position audit shows
 48 that text-input LOPO math AUC increases from 0.372 at the final layer to 0.931 at layer 22. With
 49 audio calibration and audio input, the final-layer probe does not show this failure, so we use the
 50 middle layer for its robustness across the tested conditions, not because the output layer never holds
 51 competence information. Matched controls link the late-layer degradation to the tested fine-tuned
 52 checkpoints but do not identify its cause. Second, a text-calibrated layer-22 probe transfers to speech-
 53 input states at AUC 0.855; we separate this representation transfer from the extra calibration that
 54 native serving needs. Third, a native-protocol benchmark shows that a separately fitted answer-
 55 onset gate selects useful expert calls. It raises internal answer-content accuracy by 22.1 points
 56 and exceeds the reported random-mixture reference by 6.8 points at the aggressive tier. We report
 57 accuracy together with reconstructed time to first audio, and we distinguish this benchmark from a
 58 causal streaming evaluation of audible responses.

59 2 Related Work

60 **Competence readout.** Correctness can be estimated from verbal self-evaluation [Kadavath et al.,
 61 2022, Lin et al., 2022], hidden representations [Azaria and Mitchell, 2023, Burns et al., 2023, Marks
 62 and Tegmark, 2024], and uncertainty across sampled answers [Kuhn et al., 2023, Farquhar et al.,
 63 2024]. Intermediate-layer representations can expose information missing from model outputs [Or-
 64 gad et al., 2025, Mahaut et al., 2024]; Semantic Entropy Probes amortize uncertainty estimation into
 65 a hidden-state read [Kossen et al., 2024]. Our question is how readout location, input modality, and
 66 serving protocol affect this signal after speech and duplex fine-tuning.

67 **Selective routing.** FrugalGPT, HybridLLM, and RouteLLM allocate work across language mod-
 68 els [Chen et al., 2023, Ding et al., 2024, Ong et al., 2024]; AutoMix verifies a drafted answer before

escalating [Aggarwal et al., 2024]. Pre-generation activation routers and frozen-state success probes are especially close to our method [Varshney et al., 2026, Lugoloobi et al., 2026]. A linear failure probe is thus not itself the novelty. We test its robustness in speech-model representations and use a threshold to expose a tunable expert-call budget. Predicting local failure remains distinct from predicting the benefit of a particular expert.

Native duplex control. Native spoken-dialogue models coordinate listening and speaking [Défossez et al., 2024, Cui et al., 2026], using state or synchronization mechanisms [Wang et al., 2024a, Zhang et al., 2024, Veluri et al., 2024] or frozen-backbone adapters [Wang et al., 2024b]. BayLing-Duplex [Fang et al., 2026] makes dialogue-state decisions within a single autoregressive LLM by interleaving user audio, assistant text, and assistant audio and introducing four state tokens. Starting from GLM-4-Voice, it learns turn-taking and interruption through supervised full-duplex fine-tuning and timing-focused preference optimization. Its evaluation measures spoken-answer quality and interaction timing, providing a concrete example of learned native floor control.

Competence-aware handoff. DuplexOmni, chronological thinking, and MoshiRAG connect speech to reasoning or retrieval [Huang et al., 2026, Wu et al., 2026, Chien et al., 2026]. Our work studies a decision complementary to native floor control: given a duplex-trained checkpoint, can its hidden states predict local-answer failure and guide escalation to an expert under a tunable call budget? We fit a small readout without updating the speech model. This requires regime-specific calibration and does not assume that a learned turn-taking signal predicts answer correctness.

3 Probe-Guided Expert Handoff

3.1 The local-expert loop

Our system pairs a local full-duplex speech model with a stronger external expert (Figure 1). The local model processes incoming speech and decides when to respond. At that transition, the system chooses whether to continue with the local answer or request expert help. On the expert path, an audio uplink transcribes the question, the expert returns a text answer, and the local talker voices the prepared relay text. The expert request and relay add cost and waiting time. Our goal is to improve answer accuracy under a limited expert-call budget by selecting questions that benefit from expert help. We use local-answer failure risk to guide this selection; the benefit still depends on whether the expert repairs the error and the relay preserves its answer.

3.2 A probe for the handoff decision

The local model already computes hidden states as it processes the question. If these states carry information about its ability to answer, a small classifier can read that signal without generating a separate self-evaluation. We use a linear *probe* to test whether the existing representations contain a directly readable failure signal.

Offline, we collect local answers to calibration questions and use a judge to assign correctness labels. Each label is paired with hidden-state features captured at the chosen read point. We fit a logistic probe to predict failure from these features while keeping the speech model frozen. For features $\phi(x)$ from input context x , the fitted weights w, b give the risk score

$$s(x) = \sigma(w^\top \phi(x) + b). \quad (1)$$

At inference, the probe reads the current features without a reference answer or an online judge. Scores above a language-specific threshold τ_ℓ favor escalation; thresholds are set from calibration scores and held fixed during evaluation (§4.1). An auxiliary dialogue-act head restricts escalation to information requests, filtering turns such as acknowledgments and stop commands (Appendix D).

3.3 Where should the probe read?

The probe needs a place to read the model’s internal state. The final-layer state is a natural starting point because it feeds the output head, but its ability to predict failure may change with the query distribution and input modality. Reading an earlier layer or pooling several token positions may provide a more transferable signal. Timing matters as well: a signal that requires a completed answer arrives

too late to guide the initial handoff. We therefore examine layer, position, and read time together, asking where a useful failure signal is available early in the response (§4.2–4.3; Appendix D.2).

The native system uses L22 and concatenates the latest tail state, the mean of the tail states, and the mean input state. It reads them when the generation call first returns a listen-to-speak transition. This call can already contain the first generated text unit, so we call the decision point *answer onset*. Waiting for further answer chunks may reveal additional failure information, but delays the handoff. The native probe is fitted on features and answer labels collected in this serving regime; its calibration is separate from the representation-transfer study.

4 Experiments

4.1 Experimental setup

We evaluate frozen MiniCPM-o 4.5 (9B), with gpt-5.5 as the escalation expert, on the fixed 240-query internal test set and four external English speech-QA pools: Speech TriviaQA, Speech Web Questions, and Speech Llama Questions from OpenAudioBench [Li et al., 2025], and SD-QA from VoiceBench [Faisal et al., 2021, Chen et al., 2024]. The native probe is fitted on 5,228 calibration rows; per-language thresholds use core out-of-fold scores and stay fixed across pools. Seven calibration rows overlap external questions; the recorded arms use the original fit. Calibration details, model controls, and the overlap audit are in Appendix A.

Each query has one sampled native session per arm: local-only, conservative, balanced, aggressive, and always-escalate. We report *answer-content accuracy*, scored from generated local text or intended relay text. The random reference is the analytical mixture $A_{\text{mix}}(r) = (1-r)A_{\text{local}} + rA_{\text{always}}$ at the gate’s realized call rate r . Table 1 uses the original query weights; Figure 3’s Internal curve uses the category weights specified in its caption. Both use the same recorded outcomes. External averages weight the four QA pools equally. The expert receives the final 30 seconds of the prerecorded question; timing is reconstructed from logged stages with different local and relay endpoints. These results therefore measure routing and content, without establishing causal streaming or client-audible latency. Full decoding, data, judging, and timing details are in Appendix A.1.

4.2 Layer and position matter

We first evaluate the read location independently of the expert path. A layer-by-position sweep under leave-one-pool-out (LOPO) transfer localizes the text-input failure to late last-token states (Figure 2). A last-token probe at L22 reaches **0.931** AUC on held-out math, compared with the final-layer baseline of 0.372, while mean pooling reduces the late-layer degradation. A raw Qwen backbone, subsampled to match per-pool failure rates, remains near 0.96; the Qwen2.5 control family shows a similar ordering (Table 5). These comparisons associate the degradation with the tested fine-tuned checkpoints, without isolating a causal training objective.

With audio calibration and audio input, the final-layer probe reaches 0.936 on LOPO math. We retain L22 for robustness across the tested conditions; the mechanism behind the text-input degradation remains open (Appendix B.2).

4.3 Text-to-speech representation transfer

A text-calibrated L22 probe evaluated on speech-input states reaches **0.855** AUC. Transfer stabilizes around the middle of the network, replicates on MiniCPM-o 2.6 and Qwen2.5-Omni, and is weaker in the frozen-backbone Freeze-Omni adapter control. The mid-layer pattern also appears on 200 SD-QA human recordings (Appendix B.1). These results concern representation transfer. The escalation experiments below use a separately fitted native probe with three L22 positions, as defined in §3.3.

4.4 Internal accuracy and timing cost

The upper MiniCPM block of Table 1 reports answer-content accuracy on the fixed 240-query internal test set: 40.8% locally, then 46.7%, 54.2%, and 62.9% for conservative, balanced, and aggressive. Their realized escalation rates are 8.3%, 25.0%, and 51.7% (Appendix D.1). Aggressive exceeds the

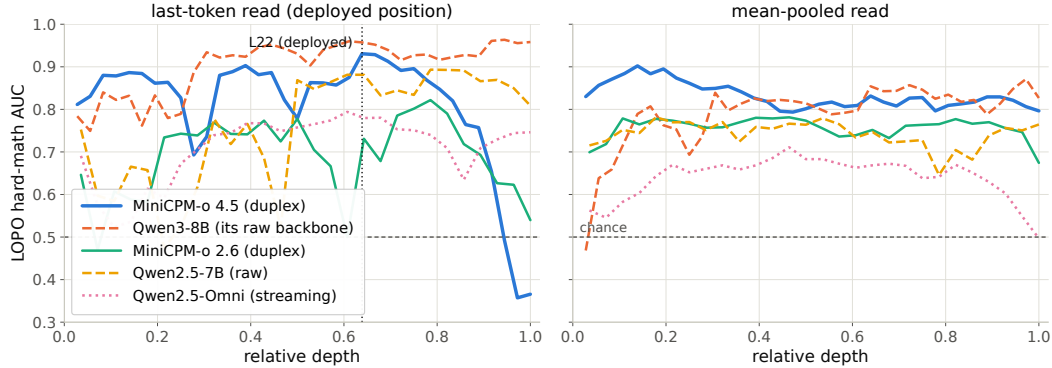


Figure 2: **Text-input transfer depends on layer and position.** LOPO math AUC across layers: late last-token reads deteriorate on the duplex checkpoints, while mid-layer reads and raw-backbone controls remain predictive. Layer 22 is the MiniCPM-o 4.5 read location used for the native gate. This plot does not show an equivalent collapse under audio calibration and audio input.

Table 1: **Main results: answer-content accuracy (%)**. Both blocks report the internal test set and four external English speech-QA pools; avg is the equal-pool external mean and Δ is its gain over always-local, computed before rounding. MiniCPM uses one recorded native session per query and arm, with fixed calibration thresholds. Matched random mixes its local and measured always arms at each pool’s realized gate rate (Appendix D.1). NVDA uses native-frame replay with cached outcomes; its Internal column is a post-selection diagnostic. Replay details and separate AlpacaEval scores are in Appendix A.1.

	Internal	TriviaQA	WebQ	Llama Q.	SD-QA	avg	Δ
<i>MiniCPM native sessions (answer-content accuracy)</i>							
always-local	40.8	62.8	52.8	82.4	48.0	61.5	—
+ gate, conservative	46.7	60.4	59.2	78.8	54.0	63.1	+1.6
+ gate, balanced	54.2	70.0	62.4	82.4	65.0	70.0	+8.5
+ gate, aggressive	62.9	88.4	75.2	85.6	82.5	82.9	+21.4
matched random, aggressive	56.1	81.5	68.9	84.1	73.7	77.1	+15.6
always-escalate (measured)	70.4	96.0	79.6	91.6	87.0	88.6	+27.1
<i>NVDA three-layer answer-onset probe (pass-3 native-frame replay)</i>							
always-local	19.6	35.8	37.6	70.8	31.0	43.8	—
+ gate @15%	31.7	48.4	48.8	80.7	42.0	55.0	+11.2
+ gate @30%	44.6	61.0	59.2	85.8	54.0	65.0	+21.2
+ gate @50%	60.4	76.8	69.6	89.7	66.5	75.7	+31.9
matched random @50%	53.5	65.7	60.2	82.0	60.0	67.0	+23.2
always-expert (ceiling)	91.7	95.5	82.8	93.1	89.0	90.1	+46.3

table’s random reference of 56.1% by 6.8 percentage points. Per-tier internal comparisons are in Table 8. Aggressive recovers 74.6% of the measured local-to-always accuracy gap while using 51.7% expert calls. This measures selection value relative to the recorded endpoints.

Gains vary across the fixed internal strata: factual QA improves by 52.5 points, while math accuracy is unchanged. Of 60 knowledge questions, 43 exceed the expert’s input window. These descriptive results identify an input constraint and task differences, without attributing the accuracy gap to truncation or changing the test mixture (Appendix A.2).

The timing diagnostic makes the cost visible (Figure 3). Internal mean time to first audio increases from 5.7 s locally to 8.7 s at balanced and 14.4 s at aggressive; aggressive P95 is 60.8 s. The always arm averages 17.6 s (P50 11.5 s). The expert call itself accounts for 6.0 s of that mean; the rest is the blocking, non-streaming synthesis of the expert text, whose tail is driven by code and markup voiced verbatim (Appendix A). A lower call rate can reduce expert-path exposure, but these tails do not support calling the current aggressive setting low latency.

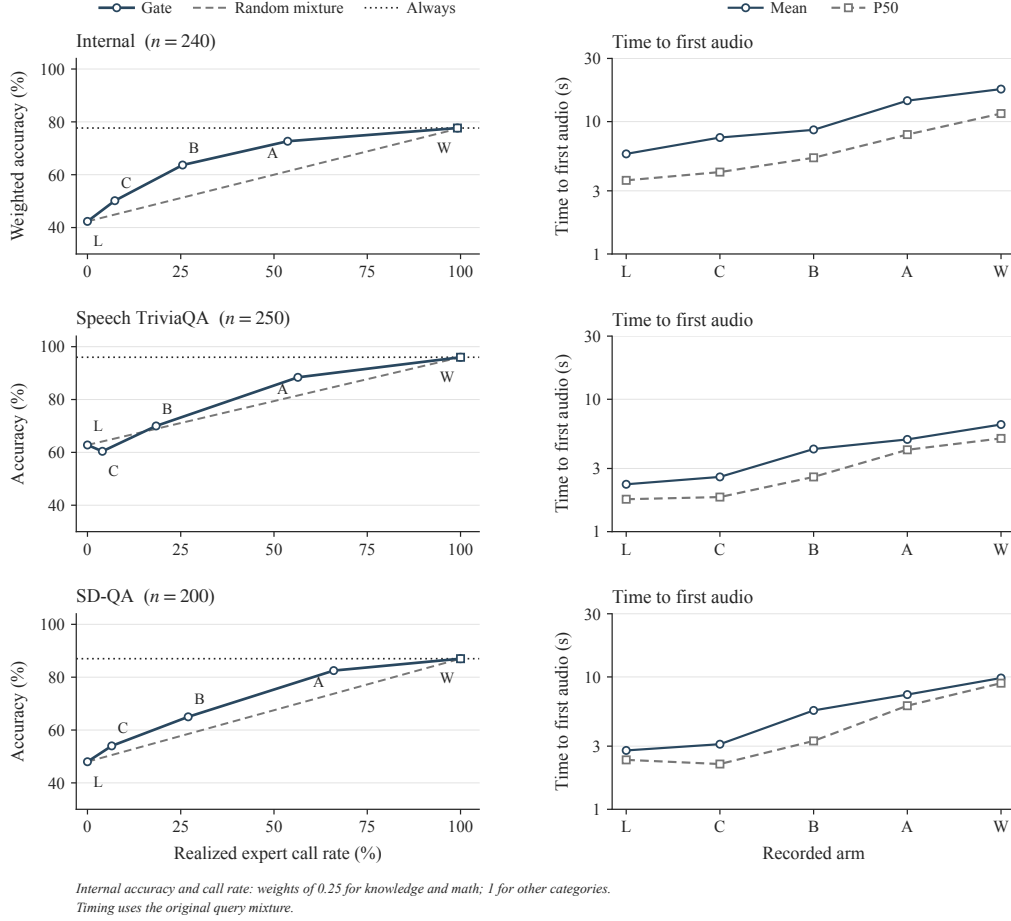


Figure 3: **Accuracy gains and timing costs on three pools.** Left: answer-content accuracy against expert call rate. The Internal curve uses post-hoc weights of 0.25 for knowledge and math and 1 for other categories, for both accuracy and call rate; all 240 queries remain included. The Internal column of Table 1 retains its original query weights. External curves are unweighted. C/B/A denote conservative/balanced/aggressive. The dashed line uses the original Table 1 random-mixture convention (§4.1). The horizontal always-arm reference passes through the measured always point; it is not an oracle bound. Right: mean and P50 of the reconstructed time to first audio on a log scale, using the original query mixture. L/W denote local/always. Each point uses the original recorded pass, with one run per query and arm; no repeated-run uncertainty is implied. Local rows end at answer-text completion and escalated rows when the relay waveform is synthesized; playback duration is excluded (§4.1). Internal P95/P99 are reported in Table 9. All four external QA pools appear in the upper block of Table 1.

4.5 External transfer and its limits

The native gate’s weights, feature definition, layer, and per-language thresholds stay fixed across the four external English QA pools. The upper block’s *avg* excludes the internal set; Δ is its gain over always-local. Aggressive macro accuracy rises from **61.5% to 82.9%**, compared with 77.1% for the reported random mixture: +21.4 points over local answering and +5.9 over random. Its realized call rate ranges from 18.8% on Llama Questions to 66.0% on SD-QA, so this is not a uniform 50%-budget comparison. TriviaQA and SD-QA illustrate large improvements over local answering (+25.6 and +34.5 points) and over the corresponding random reference (+6.9 and +8.8 points).

The lower tiers are less reliable. Conservative accuracy falls by 2.4 points on TriviaQA and 3.6 on Llama Questions; balanced leaves Llama Questions unchanged. Separately sampled sessions and limited expert headroom constrain the interpretation of small differences. The results support

187 aggressive-tier gains across these pools, not improvement on every benchmark at every budget. On
188 AlpacaEval, we measure 3.68 locally and 4.01 at aggressive, with a random-mixture reference of
189 approximately 3.9; the always arm scores 4.20 versus 4.95 for returned expert text (Appendix A.1).
190 The relay cap is particularly restrictive for long, open-ended answers.

191 Failure categories and later reads help explain the boundary. In the native failure-type audit, specific-
192 but-wrong answers comprise 70% of failures and are ranked at AUC 0.81, while execution slips in
193 multi-step reasoning are near chance (0.47). In a separate read-time sweep, external mean AUC
194 increases from 0.755 at commitment to 0.791 after two answer chunks. This is consistent with
195 some useful information emerging during generation, without proving a causal mechanism (Ap-
196 pendix D.2).

197 The lower block of Table 1 reports the exploratory NemotronLabs-VoiceChat-11B (NVDA) study.
198 Its *avg* covers four English QA pools and excludes the Internal column. At a 50% replay budget,
199 average accuracy rises from 43.8% to 75.7% ($\Delta = 31.9$ points), compared with 67.0% for random.
200 This is native-frame replay with cached outcome mixtures, not another set of MiniCPM native ses-
201 sions. The L26/L30/L34 heads reach 0.818 out-of-fold AUC and 0.816 mean external AUC in the
202 separate probe analysis (Appendix C); Table 1 reports accuracy rather than AUC. External pools
203 were inspected during variant review, so independent confirmation is needed.

204 5 Discussion and Limitations

205 The central finding is that a mid-layer read remains predictive in tested conditions where a final-
206 layer, last-token probe fails. This read location also supports selective expert calls without updating
207 the speech model.

208 Three boundaries define the scope of the results. First, coverage is limited to the tested checkpoints
209 and predominantly single-turn speech QA. Broader conversational tasks, including multi-turn dia-
210 logue and expert continuity, remain to be evaluated. Second, native serving requires in-regime audio
211 features and answer labels to fit the readout and set thresholds; representation transfer alone does not
212 provide a zero-shot gate. Third, the benchmark measures answer-content accuracy and reconstructed
213 serving time. Its long expert-path tails do not establish a latency benefit; human-rated speech quality
214 and common-clock measurements of audible responses would better capture the listener’s experi-
215 ence.

216 These results motivate competence-aware handoff as a useful addition to a speech model’s floor
217 control. Appendix A records the calibration, scoring, and input protocol needed to interpret the
218 benchmark.

References

- Pranjal Aggarwal, Aman Madaan, Ankit Anand, Srividya Pranavi Potharaju, Swaroop Mishra, Pei Zhou, Aditya Gupta, Dheeraj Rajagopal, Karthik Kappaganthu, Yiming Yang, Shyam Upadhyay, Manaal Faruqui, and Mausam. AutoMix: Automatically mixing language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2310.12963.
- Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it’s lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. arXiv:2304.13734.
- Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2212.03827.
- Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023.
- Yiming Chen, Xianghu Yue, Chen Zhang, Xiaoxue Gao, Robby T. Tan, and Haizhou Li. VoiceBench: Benchmarking LLM-based voice assistants. *arXiv preprint arXiv:2410.17196*, 2024.
- Chung-Ming Chien, Manu Orsini, Eugene Kharitonov, Neil Zeghidour, Karen Livescu, and Alexandre Défossez. MoshiRAG: Asynchronous knowledge retrieval for full-duplex speech language models. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2604.12928.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. GSM8K.
- Mike Conover, Matt Hayes, Ankit Mathur, Jianwei Xie, Jun Wan, Sam Shah, Ali Ghodsi, Patrick Wendell, Matei Zaharia, and Reynold Xin. Free dolly: Introducing the world’s first truly open instruction-tuned LLM. Databricks blog, 2023. databricks-dolly-15k.
- Junbo Cui, Bokai Xu, Chongyi Wang, Tianyu Yu, Weiyue Sun, Yingjing Xu, et al. MiniCPM-o 4.5: Towards real-time full-duplex omni-modal interaction. *arXiv preprint arXiv:2604.27393*, 2026.
- Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. Moshi: a speech-text foundation model for real-time dialogue. *arXiv preprint arXiv:2410.00037*, 2024.
- Dujian Ding, Ankur Mallick, Chi Wang, Robert Sim, Subhabrata Mukherjee, Victor Rühle, Laks V. S. Lakshmanan, and Ahmed Hassan Awadallah. Hybrid LLM: Cost-efficient and quality-aware query routing. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2404.14618.
- Fahim Faisal, Sharlina Keshava, Md Mahfuz ibn Alam, and Antonios Anastasopoulos. SD-QA: Spoken dialectal question answering for the real world. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, 2021. arXiv:2109.12072.
- Qingkai Fang, Shoutao Guo, and Yang Feng. BayLing-Duplex: Native full-duplex speech dialogue with a single autoregressive LLM. *arXiv preprint arXiv:2606.14528*, 2026.
- Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630:625–630, 2024.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks*, 2021. arXiv:2103.03874.
- Muye Huang, Lingling Zhang, Xingyu Yu, Lei Shi, Zhanyu Ma, Jun Xu, Jiuchong Gao, Jinghua Hao, Renqing He, and Jun Liu. DuplexOmni: Real-time listening, seeing, thinking, and speaking for full-duplex interaction. *arXiv preprint arXiv:2606.09186*, 2026.
- Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2017.

267 Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez,
268 Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language mod-
269 els (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

270 Jannik Kossen, Jiatong Han, Muhammed Razzak, Lisa Schut, Shreshth Malik, and Yarin Gal. Se-
271 mantic entropy probes: Robust and cheap hallucination detection in LLMs. *arXiv preprint*
272 *arXiv:2406.15927*, 2024.

273 Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for
274 uncertainty estimation in natural language generation. In *International Conference on Learning*
275 *Representations (ICLR)*, 2023. arXiv:2302.09664.

276 Tianpeng Li, Jun Liu, Tao Zhang, Yuanbo Fang, Da Pan, Mingrui Wang, et al. Baichuan-Audio:
277 A unified framework for end-to-end speech interaction. *arXiv preprint arXiv:2502.17239*, 2025.
278 Introduces OpenAudioBench.

279 Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan
280 Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International*
281 *Conference on Learning Representations (ICLR)*, 2024. arXiv:2305.20050; source of the MATH-
282 500 subset.

283 Stephanie Lin, Jacob Hilton, and Owain Evans. Teaching models to express their uncertainty in
284 words. *Transactions on Machine Learning Research*, 2022. arXiv:2205.14334.

285 William Lugoloobi, Thomas Foster, William Bankes, and Chris Russell. LLMs encode their failures:
286 Predicting success from pre-generation activations. *arXiv preprint arXiv:2602.09924*, 2026.

287 Matéo Mahaut, Laura Aina, Paula Czarnowska, Momchil Hardalov, Thomas Müller, and Lluís
288 Màrquez. Factual confidence of LLMs: on reliability and robustness of current estimators. In
289 *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.

290 Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language
291 model representations of true/false datasets. In *Conference on Language Modeling (COLM)*, 2024.
292 arXiv:2310.06824.

293 Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez,
294 M. Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs with preference data.
295 *arXiv preprint arXiv:2406.18665*, 2024.

296 Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, Hadas Kotek, and
297 Yonatan Belinkov. LLMs know more than they show: On the intrinsic representation of
298 LLM hallucinations. In *International Conference on Learning Representations (ICLR)*, 2025.
299 arXiv:2410.02707.

300 Tanya Stivers, N. J. Enfield, Penelope Brown, Christina Englert, Makoto Hayashi, Trine Heinemann,
301 Gertie Hoymann, Federico Rossano, Jan Peter de Ruiter, Kyung-Eun Yoon, and Stephen C. Levin-
302 son. Universals and cultural variation in turn-taking in conversation. *Proceedings of the National*
303 *Academy of Sciences*, 106(26):10587–10592, 2009.

304 Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy
305 Liang, and Tatsunori B. Hashimoto. Stanford alpaca: An instruction-following LLaMA model.
306 GitHub repository, 2023. zh split: shibing624/alpaca-zh.

307 Tanay Varshney, Annie Surla, Michelle Xu, Gomathy Venkata Krishnan, Maximilian Jeblick, David
308 Austin, Neal Vaidya, and Davide Onofrio. LLM router: Rethinking routing with prefix activations.
309 *arXiv preprint arXiv:2603.20895*, 2026.

310 Bandhav Veluri, Benjamin N. Peloquin, Bokai Yu, Hongyu Gong, and Shyamnath Gollakota. Be-
311 yond turn-based interfaces: Synchronous LLMs as full-duplex dialogue agents. In *Proceedings*
312 *of EMNLP (Main)*, 2024. arXiv:2409.15594.

313 Peng Wang, Songshuo Lu, Yaohua Tang, Sijie Yan, Wei Xia, and Yuanjun Xiong. A full-duplex
314 speech dialogue scheme based on large language models. In *Advances in Neural Information*
315 *Processing Systems (NeurIPS)*, 2024a. arXiv:2405.19487.

- 316 Xiong Wang, Yangze Li, Chaoyou Fu, Yunhang Shen, Lei Xie, Ke Li, Xing Sun, and Long Ma.
 317 Freeze-Omni: A smart and low latency speech-to-speech dialogue model with frozen LLM. *arXiv*
 318 *preprint arXiv:2411.00774v5*, 2024b.
- 319 Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, et al. MMLU-Pro: A more robust and
 320 challenging multi-task language understanding benchmark. In *Advances in Neural Information*
 321 *Processing Systems (NeurIPS), Datasets and Benchmarks*, 2024c. arXiv:2406.01574.
- 322 Jason Wei, Nguyen Karina, Hyung Won Chung, Yunxin Joy Jiao, Spencer Papay, Amelia Glaese,
 323 John Schulman, and William Fedus. Measuring short-form factuality in large language models.
 324 *arXiv preprint arXiv:2411.04368*, 2024. SimpleQA.
- 325 Donghang Wu, Haoyang Zhang, Chen Chen, Tianyu Zhang, Fei Tian, Xuerui Yang, Gang Yu, Hexin
 326 Liu, Nana Hou, Yuchen Hu, and Eng Siong Chng. Chronological thinking in full-duplex spoken
 327 dialogue language models. In *Proceedings of the 27th Annual Meeting of the Special Interest*
 328 *Group on Discourse and Dialogue (SIGDIAL)*, 2026. arXiv:2510.05150.
- 329 Xinrong Zhang, Yingfa Chen, Shengding Hu, Xu Han, Zihang Xu, Yuanwei Xu, Weilin Zhao,
 330 Maosong Sun, and Zhiyuan Liu. Beyond the turn-based game: Enabling real-time conversations
 331 with duplex models. In *Proceedings of EMNLP*, 2024. arXiv:2406.15718.

A Benchmark and calibration details

The main table reports the original internal query mixture and four external English QA pools. The figure’s Internal accuracy and call-rate curve uses post-hoc knowledge/math weights of 0.25 and weights of 1 for other categories; timing retains the original mixture. Archived Mandarin diagnostics are separate from the main-table macro average.

A.1 Experimental setup details

Models, data, and calibration. The target is MiniCPM-o 4.5 (9B, based on Qwen3-8B). Matched controls include raw backbones, a second MiniCPM generation, Qwen2.5-Omni, Qwen2-Audio, and Freeze-Omni (Table 3). Representation analyses use bf16, greedy decoding, and seed 42. The native benchmark uses sampled decoding under the serving configuration ($T = 0.7$, $\text{top-}k = 20$); these are different experimental regimes. The escalation expert is gpt-5.5 with low reasoning effort.

The internal 600-query set contains SimpleQA traps [Wei et al., 2024], MMLU-Pro knowledge questions [Wang et al., 2024c], TriviaQA facts [Joshi et al., 2017], Dolly/Alpaca chat prompts [Conover et al., 2023, Taori et al., 2023], and GSM8K/MATH problems [Cobbe et al., 2021, Hendrycks et al., 2021, Lightman et al., 2024]. Its original 360/240 split and test composition remain fixed. LLM judges score answer adequacy. A stronger re-judge of the representation-study labels agrees on 96.2% of text and 94.5% of audio cases; this agreement is not a validation of speech delivery.

Three calibration regimes must be distinguished: 360 examples for single-position representation probes; 2,310 for the earlier turn-based three-position gate; and 5,228 for the native gate, refitted on native features and judged native-answer outcomes. The native fit contains 4,986 core and 242 fresh training rows; thresholds use five-fold out-of-fold scores from the core rows, grouped by language. Conservative, balanced, and aggressive target nominal calibration rates of 15/30/50%; realized rates on new pools can differ. Neither fresh-row scores nor external evaluation-pool quantiles enter these thresholds. Seven fitting rows overlap external questions. A 5,221-row deduplicated artifact is available, but was not used in the recorded benchmark; its reported offline effects are small (Appendix A).

Native benchmark protocol and metrics. The MiniCPM upper block of Table 1 evaluates the internal test split and four external English speech-QA pools: Speech TriviaQA, Speech Web Questions, and Speech Llama Questions from OpenAudioBench [Li et al., 2025], plus SD-QA recordings from VoiceBench [Faisal et al., 2021, Chen et al., 2024]. VoiceBench AlpacaEval is reported separately on its 1–5 scale. The archived Mandarin Speech Reasoning QA pool is retained in the appendix diagnostics and excluded from the main-table macro average. External audio comes from the benchmark releases; internal questions use a fixed TTS rendering. Each query has a separate native-protocol session for each arm: local-only, three gate tiers, and always-escalate. There is one recorded run per query and arm, without repeated-run averaging.

Audio enters the local model in one-second chunks. At an escalation, transcription and the expert produce a text answer, which is cleaned and capped at 400 characters for teacher-forced TTS relay. We report *answer-content accuracy*: the judge sees local generated text or the intended relay text, not a transcription of played audio. The benchmark disables local audio generation. Uplink transcription uses the final 30 seconds of the full prerecorded question: it can access unconsumed audio at early onset and loses the beginning of longer questions. Thus the experiment evaluates routing under this input protocol, without establishing strictly causal interaction.

For the MiniCPM block, the reported random reference is

$$A_{\text{mix}}(r) = (1 - r)A_{\text{local}} + rA_{\text{always}},$$

where r is the gate’s realized escalation rate. This follows Table 1’s convention of treating the always policy as the full-escalation endpoint; it is an analytical mixture, not a separately run randomized arm. Non-commits and a sensitivity check using the actual always-arm rate are reported in Appendix A. Timing is reconstructed from chunk offsets and logged stage durations as time to first audio: the relay path ends when its synthesized waveform is ready (expert wait plus blocking TTS), excluding the waveform’s playback duration; the local path ends at text generation. Figure 3 reports mean and P50 of this diagnostic; Table 9 retains internal P95/P99. These summaries do not measure client-audible completion (Appendix A).

Main-table replay and open-ended scores. The NVDA lower block re-mixes local and cached expert outcomes after pass-3 replay through the native frame protocol (Appendix C). External columns retain cached local outcomes and the prior 2,481-row fit. The Internal column judges the same pass-3 answers on all 240 queries and uses a 2,258-row refit that excludes these test IDs, with cached pass-2 training labels. All policies leave the 17 queries without an answer-onset read local, so 15/30/50% scoreable-query budgets realize 14/28/47% overall. This Internal result is post-selection because probe architecture selection included this split.

AlpacaEval uses a separate 1–5 scale: MiniCPM changes from 3.68 locally to 4.01 at 44% escalation (rounded-anchor matched random 3.91; always-escalate 4.20, compared with 4.95 for returned expert text). The pass-3 NVDA ranking changes from 3.55 to 4.51 versus 4.23 for random. These open-ended scores do not enter the QA macro average.

Calibration. Representation analyses use a fixed 360-query calibration split and single-position 4,096-dimensional features. The native gate uses 5,228 rows with native-answer labels and three L22 summaries: the last tail state, the tail mean, and the input mean. The logistic head uses $C = 3 \times 10^{-4}$. Five-fold stratified cross-validation uses seed 42; thresholds are language-specific 15/30/50% quantiles of the 4,986 core rows’ out-of-fold scores. The 242 fresh rows enter fitting but not the quantile base. Exported scores reproduce thresholds to four decimals. The seven-row overlap exclusion changes the reported held-out AUCs by at most 0.002 and offline accuracy remixes by less than 0.5 point. These checks concern a separate refit; the recorded arms use the original gate and retain those overlaps. Layer selection used internal LOPO results without nested selection within each held-out pool.

Scoring and timing. Each arm contains one native-protocol session per query. Answer-content accuracy is judged from local generated text or the intended expert relay text. The expert receives the last 30 seconds of the full question waveform. Of 240 internal questions, 46 exceed that window; the always arm escalates 45 of these and drops their beginnings. The input includes not-yet-consumed audio for 18/240 aggressive sessions. Time to first audio is reconstructed from the onset chunk offset and generation, waiting, and synthesis times; relay playback duration is excluded. Local rows end at text generation, while escalated rows end when blocking TTS returns a waveform. The always arm fires on 238/240 queries. Adjusting the random reference for its 99.2% call rate changes the aggressive reference from 56.1 to 56.2%, leaving a 6.7-point gate advantage.

Repeated and audio-output checks. Two additional internal passes put aggressive accuracy in the range 62.9–65.4% (mean 64.4%, sample standard deviation 1.3 points). A separate 299-query validation uses only consumed input audio and saves both output waveforms. Local/always text accuracy is 50.8/77.3%, while accuracy after output-audio transcription is 48.2/66.2%. The validation uses a separate 4,922-row gate fit excluding its 299 queries and reports endpoint arms, rather than live gate-arm results. The repeats describe internal run variability; the transcription scores motivate direct speech evaluation.

A.2 Internal task strata

Table 2: Descriptive strata of the fixed internal test set. Local, aggressive (A), and always report answer-content accuracy (%); call rate is the aggressive arm’s realized rate. The final column counts question waveforms longer than the expert’s 30-second input window. One recorded run per query and arm; no samples are removed or reweighted.

Stratum	n	Local	A	Always	A call rate	> 30 s (n)
Chat	60	65.0	70.0	70.0	35.0	0
Factual QA	40	32.5	85.0	92.5	62.5	0
Knowledge	60	15.0	31.7	38.3	71.7	43
Math	60	61.7	61.7	78.3	25.0	3
SimpleQA traps	20	0.0	95.0	100.0	100.0	0

The largest gains occur on factual QA and SimpleQA traps. Math is unchanged at the aggregate level, consistent with the failure-type analysis in Appendix D.2. Long-input truncation is concentrated in the knowledge stratum (43 of 60 questions); all queries remain in the reported denominator.

423 B Models and representation analyses

Table 3: Model roster. “Pair” identifies a fine-tuned model and its raw backbone control.

model	type	pair	role
MiniCPM-o 4.5	omni + duplex FT	Qwen3-8B	system target
Qwen3-8B	raw	—	pair control
MiniCPM-o 2.6	duplex FT	Qwen2.5-7B	replication
Qwen2.5-Omni-7B	streaming omni FT	Qwen2.5-7B	omni control
Qwen2.5-7B	raw	—	pair control
Qwen2-Audio-7B	audio FT	Qwen2-7B	audio control
Freeze-Omni	frozen LLM + adapter	Qwen2-7B	adapter control
Qwen2-7B	raw	—	pair control

424 B.1 Signal comparison and layer transfer

Table 4: Failure-prediction AUC on MiniCPM-o 4.5. In-dist uses five-fold out-of-fold predictions; LOPO holds out the named pool.

signal	in-dist	LOPO math	audio input
max entropy@4	0.696	near chance	—
$p(\text{True})$ -pre	0.807	trap 0.945	degrades
$p(\text{True})$ -post	0.899	—	degrades
final-layer probe	0.822	0.372	0.936
mid-layer (L22) probe	0.866	0.931	0.855

425 The final-layer state predicts source pool at 95.8% accuracy, and a pool-identity oracle alone reaches
 426 failure AUC 0.715. This motivates LOPO rather than an in-distribution comparison. The depth
 427 sweep shows that the failure is localized: mid-layer last-token reads transfer while late reads degrade
 428 on both duplex models; mean pooling degrades less. In the signal table, the final-layer audio entry
 429 is audio-calibrated LOPO math, whereas the L22 audio entry tests a text-calibrated probe on speech
 430 states.

Table 5: LOPO math AUC by depth for matched model families.

model	best mid-layer	final layer	shape
Qwen3-8B	0.964 (L33/36)	0.958	plateau
MiniCPM-o 4.5	0.931 (L22/36)	0.366	late inversion
Qwen2.5-7B	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni	0.794 (L16/28)	0.746	diffuse decline
MiniCPM-o 2.6	0.822 (L21/28)	0.540	late cliff

431 The text-to-audio result is not explained by synthetic speech alone: on 200 human SD-QA record-
 432 ings, the text-calibrated read remains at 0.76–0.80 AUC across L22–L26. Audio-input verbal self-
 433 evaluation is less reliable, whereas adding a transcript restores it; the hidden-state probe does not
 434 require that extra generation step.

435 B.2 Mechanism checks

436 Matched controls narrow the late-layer effect without assigning it to a specific training objective.
 437 A speak-mode template leaves final-layer LOPO math unchanged (0.362 versus 0.366), the audio-
 438 text direction is nearly orthogonal to the failing probe, and source-pool identity is decodable at
 439 all depths. A residual-component analysis shows the distributed read holding through the middle
 440 layers and decaying only near the outputs of the two duplex models; it remains intact in their raw
 441 backbones and the omni-streaming control. This supports the practical choice of a mid-layer read
 442 while leaving the causal mechanism open.

443 B.3 Query-surface baseline

444 A TF-IDF logistic router reaches 0.669 out-of-fold AUC and 0.38–0.57 under LOPO, including
 445 0.230 on the all-failure trap pool. The hidden-state probe’s advantage is therefore not explained by
 446 query wording or source identity alone. On audio labels the surface router improves to 0.814 but
 447 still trails the L22 probe at 0.879.

448 C Second duplex family

449 We replay NemotronLabs-VoiceChat-11B through its native frame protocol with the checkpoint
 450 frozen. Of 2,550 English calibration queries, 2,481 produce a commit frame. Standardized logistic
 451 heads read L26/L30/L34 onset states; their normalized logits are averaged. The recipe is chosen
 452 on calibration-only LOPO metrics, and external labels are excluded from fitting. Each head uses
 453 $C = 10^{-4}$ and concatenates the commit state, eighth post-commit state, first-eight-state mean, and
 454 running pre-commit mean (17,920 dimensions per layer). External pools were inspected during
 455 variant review, so these results provide exploratory transfer evidence. The read covers the first 640
 456 ms after commit, so this experiment tests method transfer rather than end-to-end real-time serving.

Table 6: VoiceChat-11B transfer. Mean accuracy equally weights four English speech-QA pools; random is the matched-rate mixture.

expert budget	15%	30%	50%
probe-guided accuracy	55.0	65.0	75.7
random-mixture accuracy	50.7	57.7	67.0

457 The three-layer read reaches 0.818 out-of-fold AUC and 0.816 mean external AUC under the pool-
 458 aligned judges. A strictly causal commit-only variant loses about 0.03 external AUC, making answer
 459 onset the closest matching read point across the two model families.

460 D Native gate and serving details

461 The L22 hidden state is available after roughly 57–61% of prefill. On one H100, probe availability
 462 has P50/P95 of 20/25 ms for text and 45/104 ms for audio. These numbers measure the local read,
 463 not the network or expert call. The native benchmark’s realized rates are below; their variation is
 464 expected because thresholds are calibration quantiles rather than per-pool quotas.

465 **Dialogue-act filtering.** A second linear head uses the same L22 read to distinguish information
 466 requests from floor-management turns. Escalation requires both the information-seeking score $a(x)$
 467 and failure score $s(x)$ to exceed their respective thresholds:

$$d(x) = \mathbf{1}\{s(x) \geq \tau_\ell\} \mathbf{1}\{a(x) \geq \tau_a\}. \quad (2)$$

468 On the scripted calibration stimuli, the act head has OOF AUC 1.000; the question-side 0.5th-
 469 percentile threshold removes 0.5% of true escalations and admits no floor-management stimuli.

470 D.1 Realized escalation rates

Table 7: Realized escalation rates (%) for the MiniCPM native results in the upper block of Table 1, plus the archived Mandarin and separate AlpacaEval pools. These are measured rates, distinct from the nominal calibration budgets and from the NVDA replay budgets. Local-only makes no expert calls.

Pool	n	Conservative	Balanced	Aggressive	Always
Internal	240	8.3	25.0	51.7	99.2
Speech TriviaQA	250	4.0	18.4	56.4	100.0
Speech WebQ	250	4.8	22.0	60.0	99.2
Speech Llama Q.	250	0.4	3.6	18.8	100.0
SD-QA	200	6.5	27.0	66.0	100.0
Reasoning QA (zh)	202	19.8	33.2	50.5	100.0
AlpacaEval	199	1.5	4.5	43.7	100.0

Table 8: Internal native-protocol answer-content accuracy with query-bootstrap 95% confidence intervals ($n = 240$). Random follows Table 1’s endpoint-mixture convention.

tier	realized calls	accuracy	random reference
local-only	0.0%	.408 [.35,.47]	.408
conservative	8.3%	.467 [.40,.53]	.433
balanced	25.0%	.542 [.48,.60]	.482
aggressive	51.7%	.629 [.57,.69]	.561
always	99.2%	.704 [.65,.76]	.704

Table 9: Internal reconstructed time to first audio (seconds, $n = 240$ per arm). Local rows end at text completion and escalated rows when blocking TTS returns. Playback duration is excluded. Quantiles describe variation across queries within one run.

tier	mean	P50	P95	P99
local-only	5.7	3.6	17.3	21.2
conservative	7.6	4.2	17.5	80.2
balanced	8.7	5.3	22.9	89.6
aggressive	14.4	8.0	60.8	131
always	17.6	11.5	69.4	117

471 D.2 Failure type and later reads

472 This separate sweep fits a probe at each read point using 5,236 judged calibration rows, row-random
 473 five-fold fitting, and $C = 3 \times 10^{-4}$. Later reads pad finished answers with the last available vector.
 474 Its five-pool macro includes Mandarin reasoning, unlike the main table’s four-English-pool macro.
 475 Failure categories are model judgments.

Failure-prediction AUC at successive native read points. External macro equally weights five speech-QA pools.

Read point	Calibration OOF	Internal	Reasoning-zh	External macro
Before onset	.848	.832	.711	.753
At onset	.850	.852	.648	.755
After 1 chunk	.852	.854	.681	.762
After 2 chunks	.855	.846	.769	.791
After 3 chunks	.857	.858	.744	.801

477 Specific-but-wrong answers account for about 70% of failures and are ranked at AUC 0.813; per-
 478 ception errors reach 0.889, while execution slips are near chance at 0.469. The later-read gains on
 479 reasoning expose a tradeoff between earlier routing and information that emerges during generation.

480 **E Judge prompt and reproducibility**

481 The answer judge receives the question, an optional reference, and a candidate answer without
482 the system identity. It marks the candidate adequate only when it is factually correct, uses valid
483 reasoning, and completely answers the question; partial, hedged, refused, off-topic, and truncated
484 answers are inadequate. The pre-answer self-evaluation asks the model not to answer the question
485 and to reply exactly “Yes” or “No” to whether it would answer correctly. The post-draft version
486 presents the question and proposed answer and asks whether the answer is correct.

487 Representation-study sampling and splits use seed 42. The native benchmark uses sampled decoding
488 and retains one run per query and arm. MiniCPM-o 4.5 runs with PyTorch 2.8.0 and Transformers
489 4.51.0 on single-H100 instances. The release includes the calibration manifest, out-of-fold scores,
490 thresholds, overlap exclusions, benchmark hashes, and scripts that rebuild the figures and metric
491 summaries from judged rows.